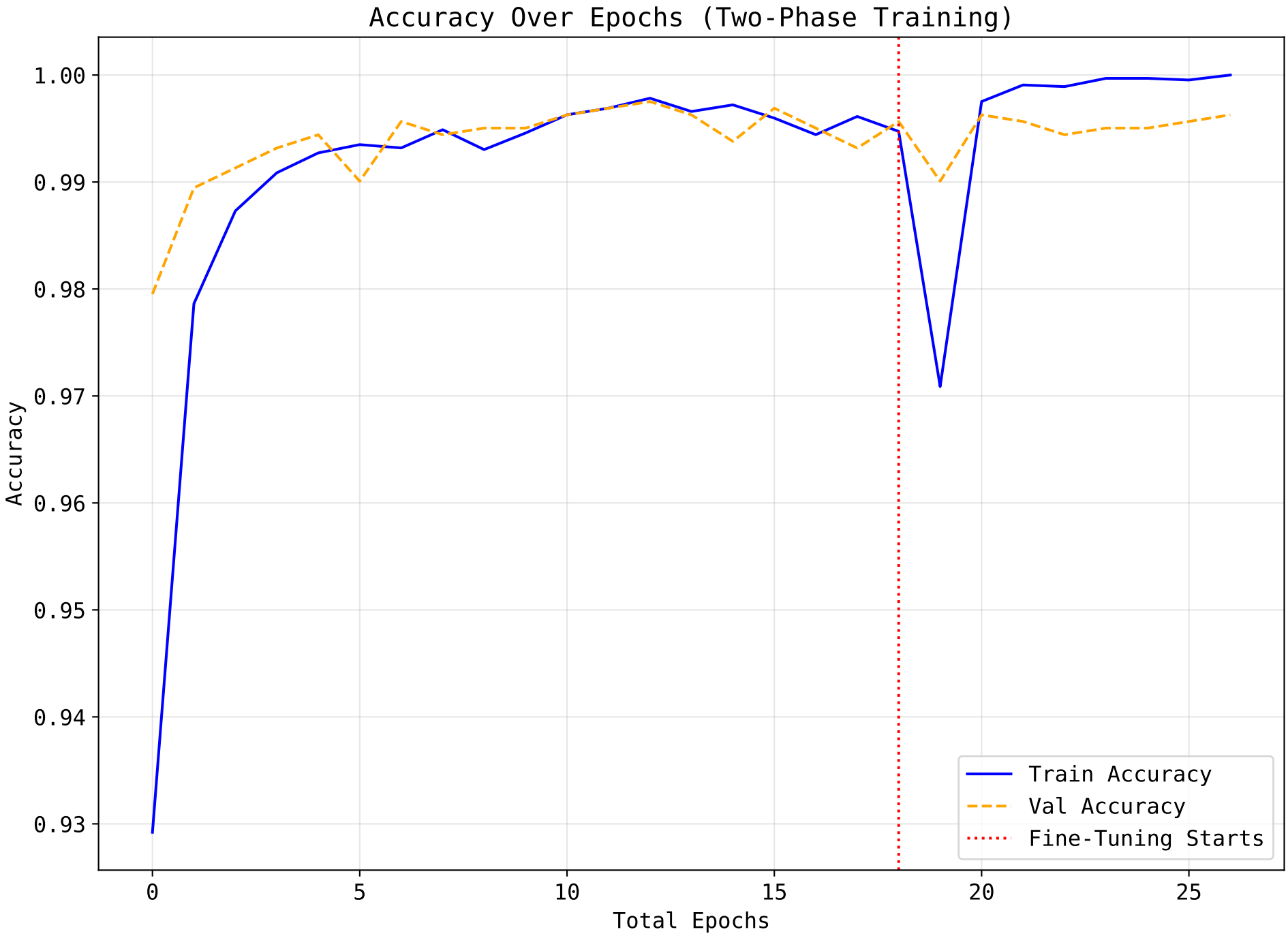
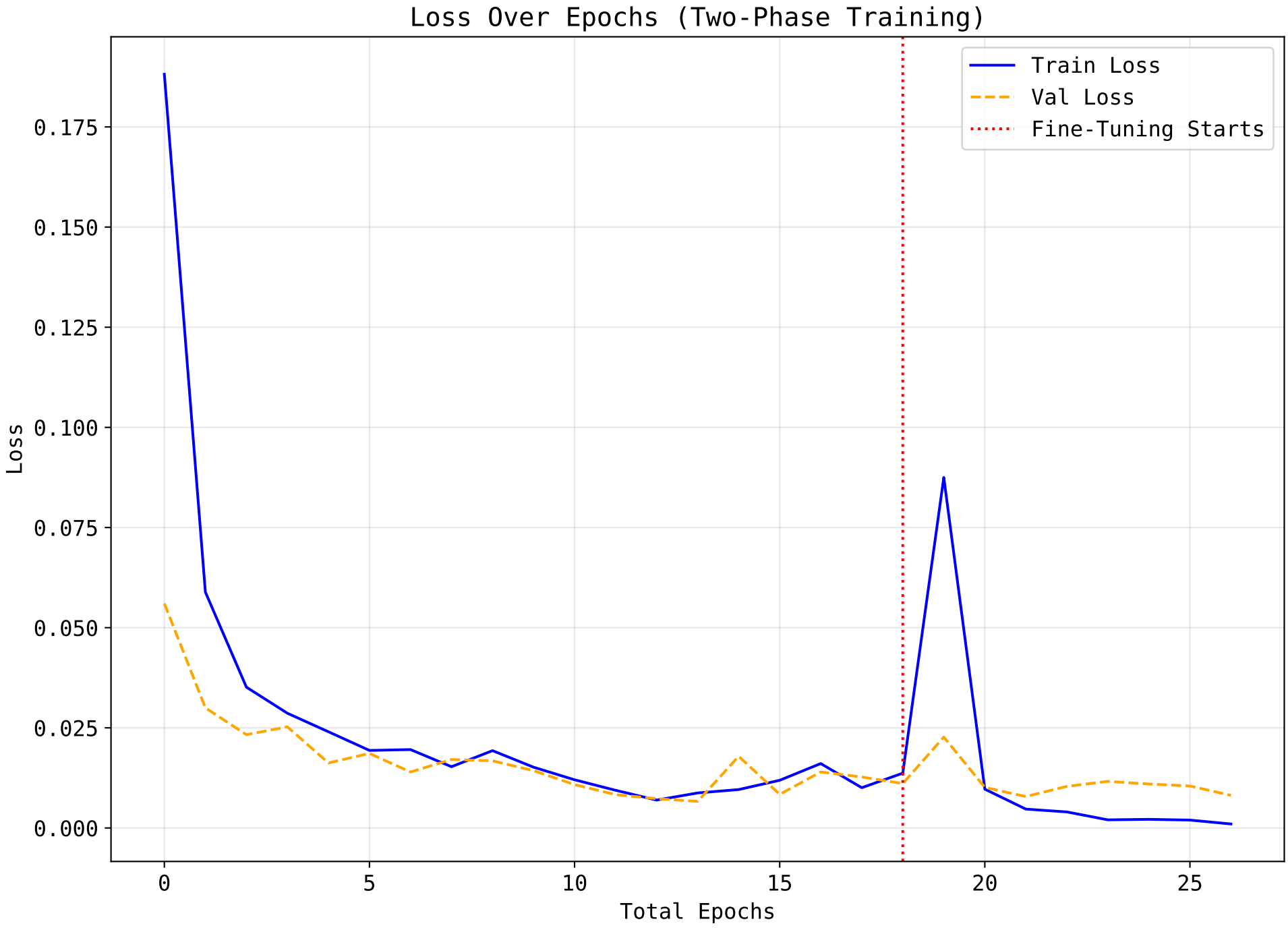
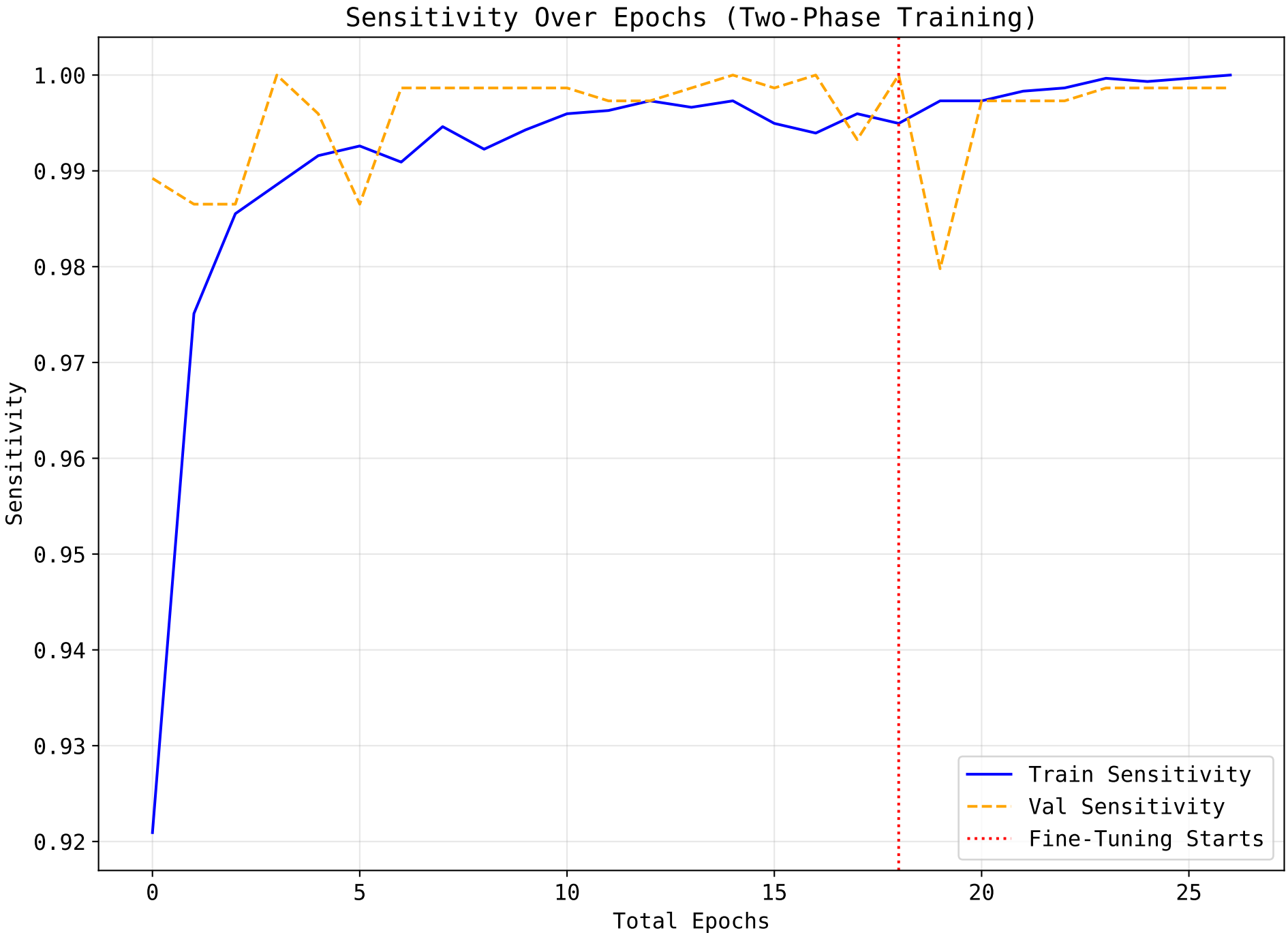
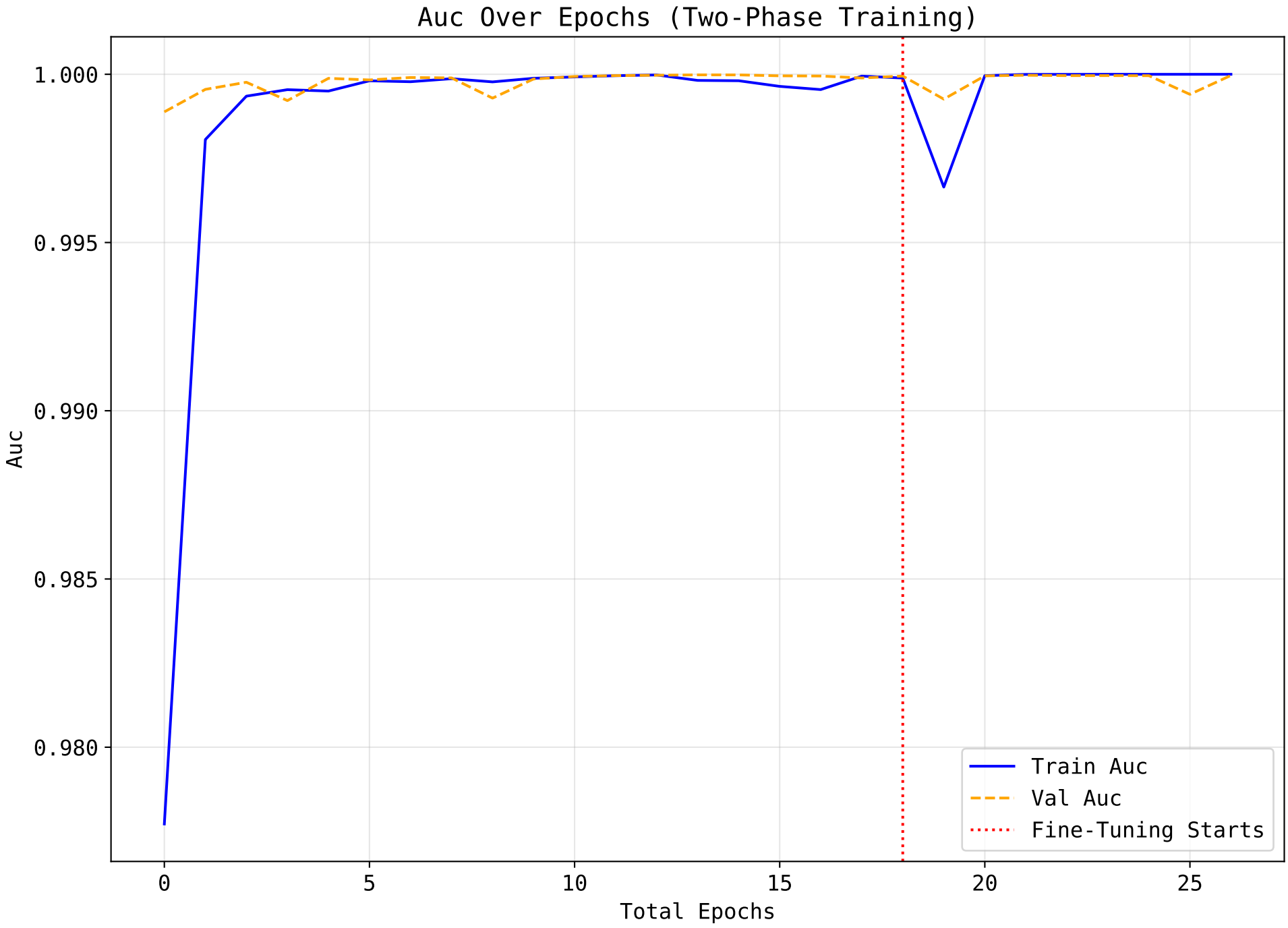
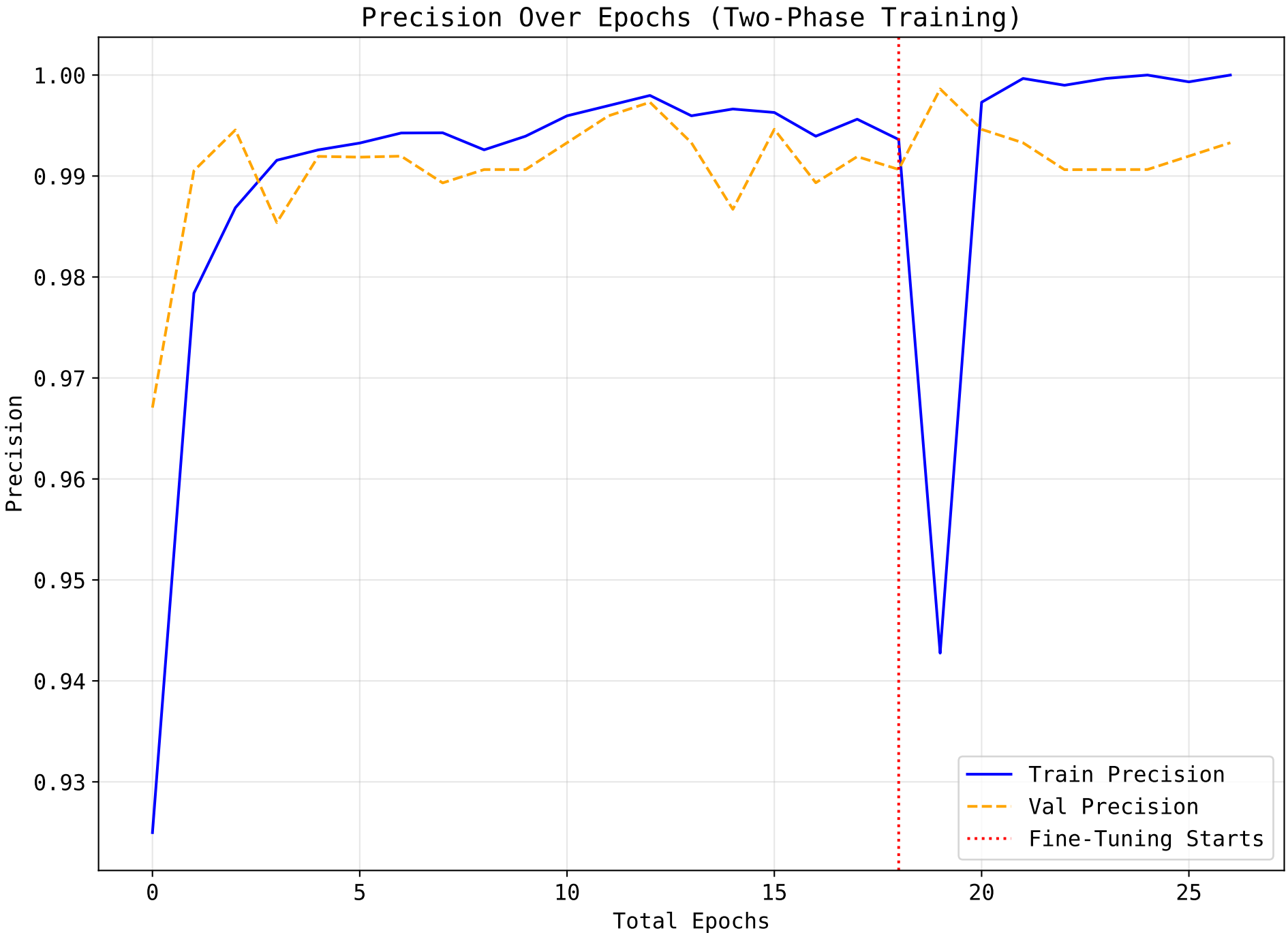
```
Model Evaluation Report
Started: 2026-04-12 17:03:04
Finished: 2026-04-12 17:18:04

Model Architecture: ResNet50
Early Stopping Patience: 5
Phase 1 Learning Rate: 0.001
Phase 2 Learning Rate: 1e-05
Phase 1 Epochs:           19 (Limit: 30)
Phase 2 Epochs:           8 (Limit: 50)
Class Weights: {0: 1.0, 1: 1.0}
Cropped training dataset

Test Loss                : 0.5603
Binary Accuracy          : 0.8803
Precision                : 0.9024
Sensitivity (Recall)     : 0.8520
AUC                      : 0.9369
Calculated Specificity: 0.9085
Classification Report:
              precision    recall  f1-score   support

      Normal       0.86      0.91      0.88       306
    Cataract       0.90      0.85      0.88       304

    accuracy                           0.88       610
   macro avg       0.88      0.88      0.88       610
weighted avg       0.88      0.88      0.88       610
```

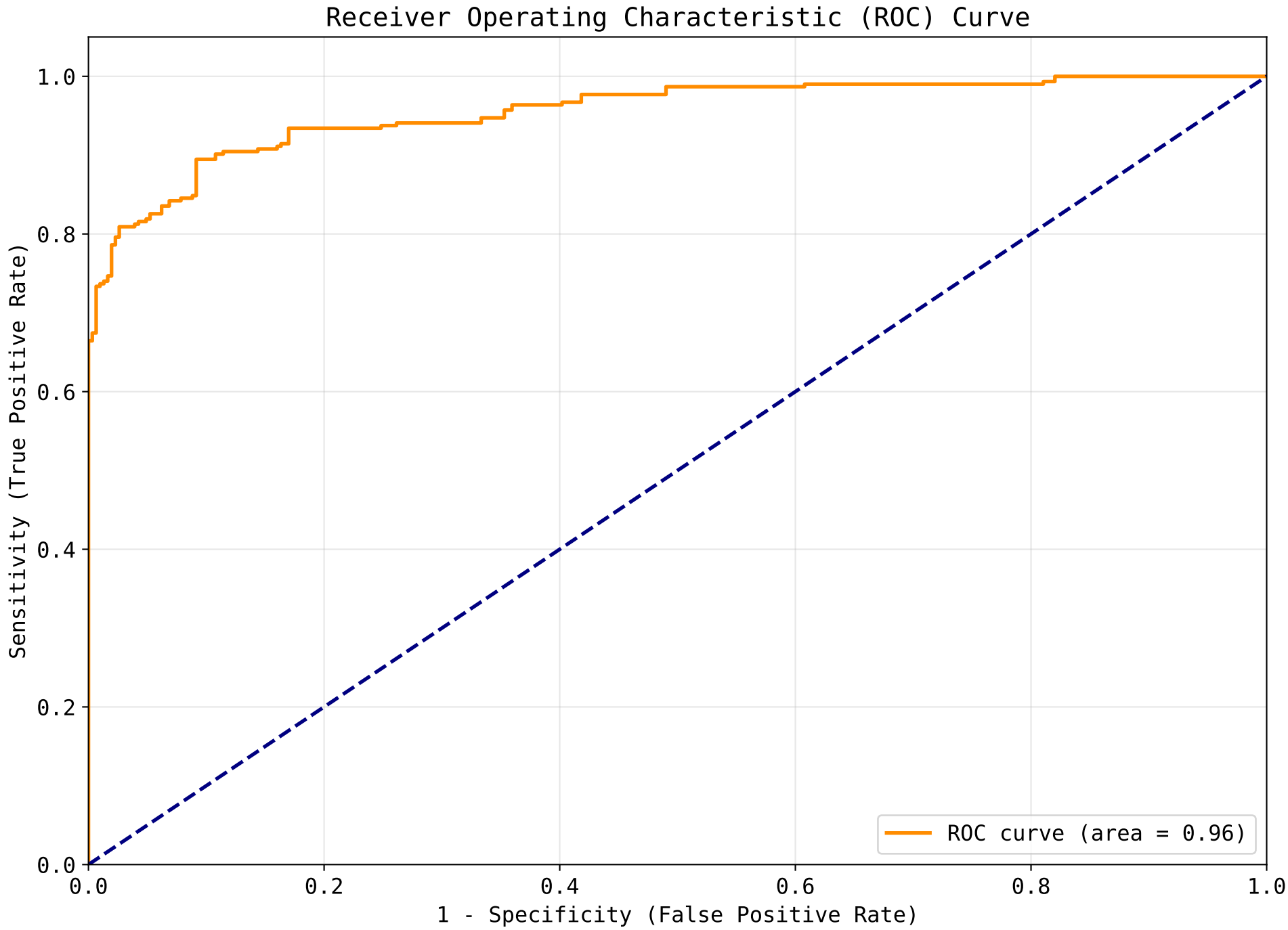

Accuracy Over Epochs (Two-Phase Training)
Accuracy
Total Epochs
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0
5
10
15
20
25
Train Accuracy
Val Accuracy
Fine-Tuning Starts

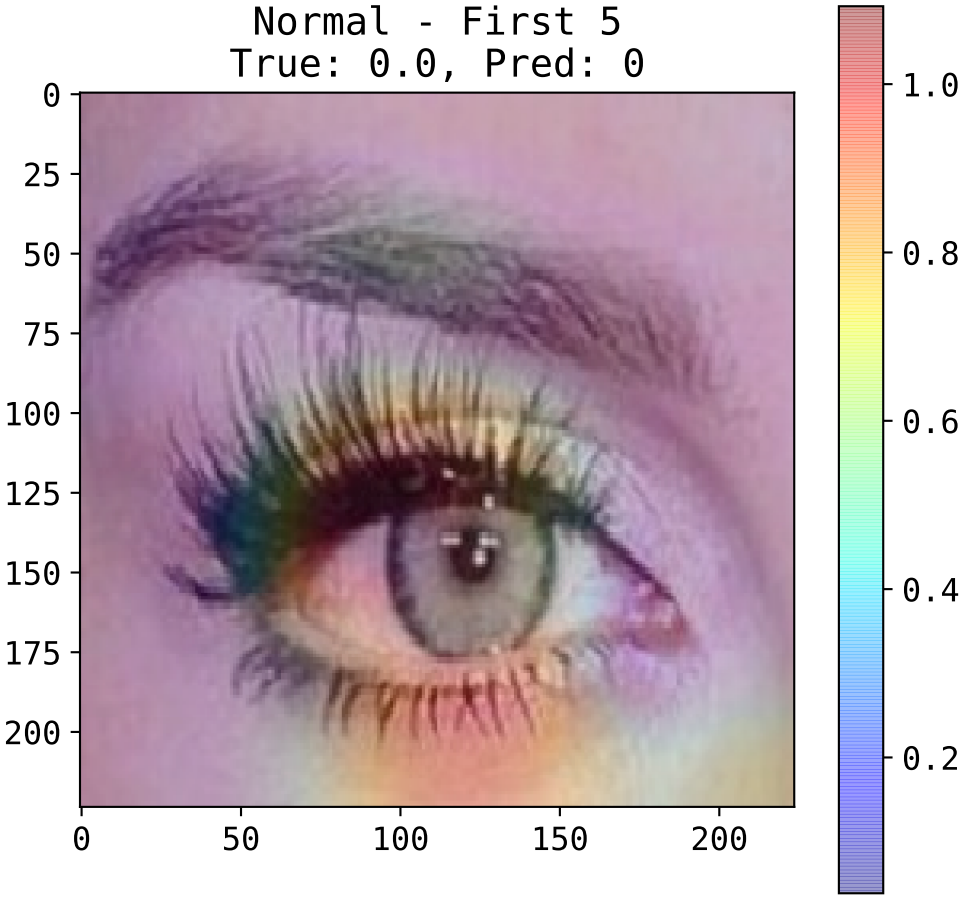

Loss Over Epochs (Two-Phase Training)
Train Loss
Val Loss
Fine-Tuning Starts
Loss
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
5
10
15
20
25
Total Epochs

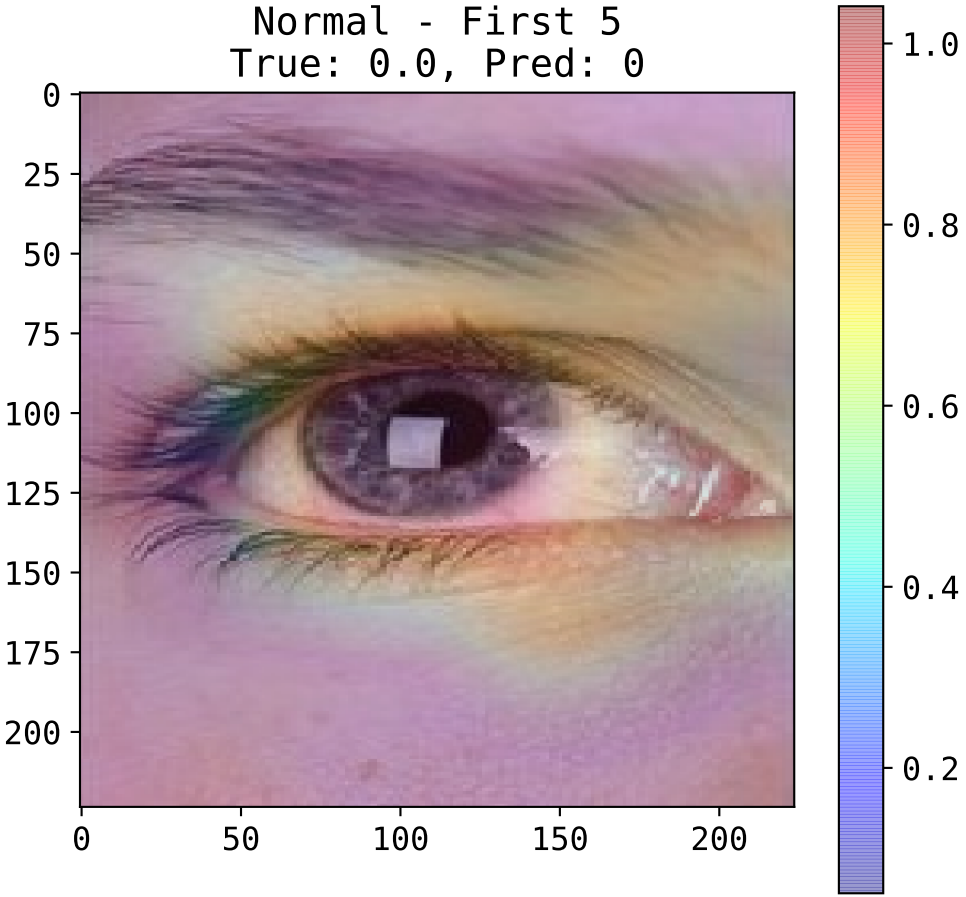

Sensitivity Over Epochs (Two-Phase Training)
Sensitivity
Total Epochs
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0.92
0
5
10
15
20
25
Train Sensitivity
Val Sensitivity
Fine-Tuning Starts

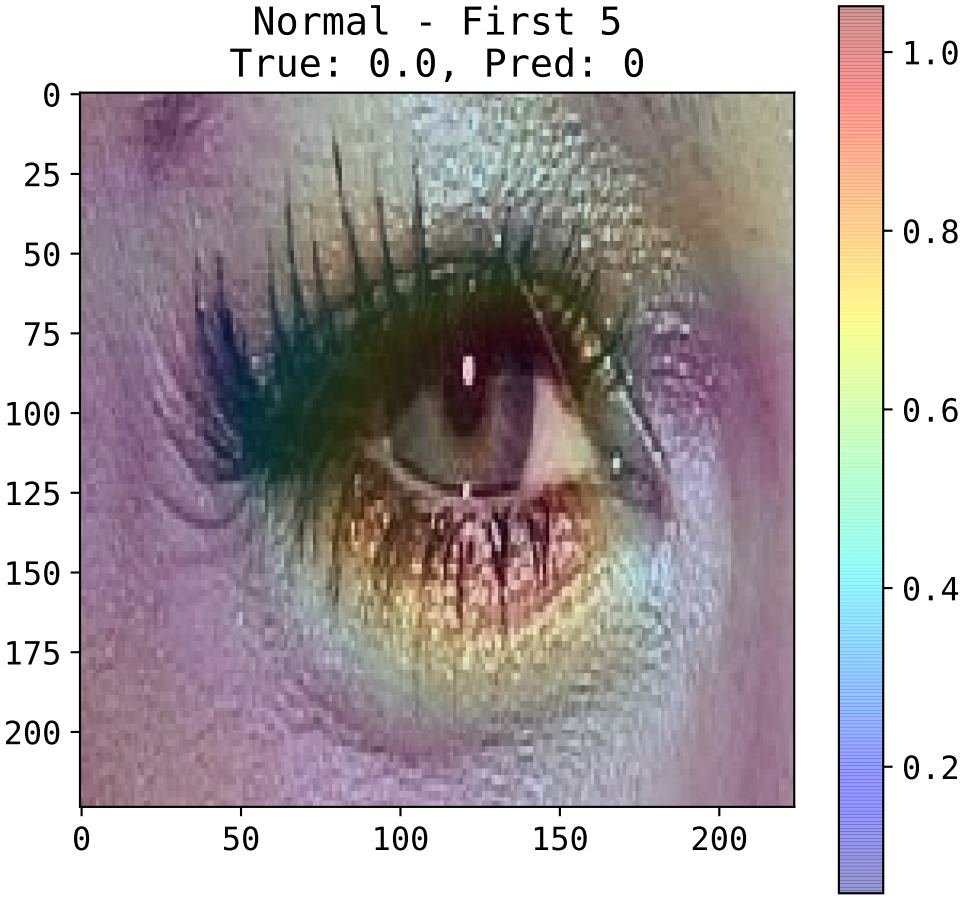

Auc Over Epochs (Two-Phase Training)
1.000
0.995
0.990
0.985
0.980
Auc
0
5
10
15
20
25
Total Epochs
Train Auc
Val Auc
Fine-Tuning Starts

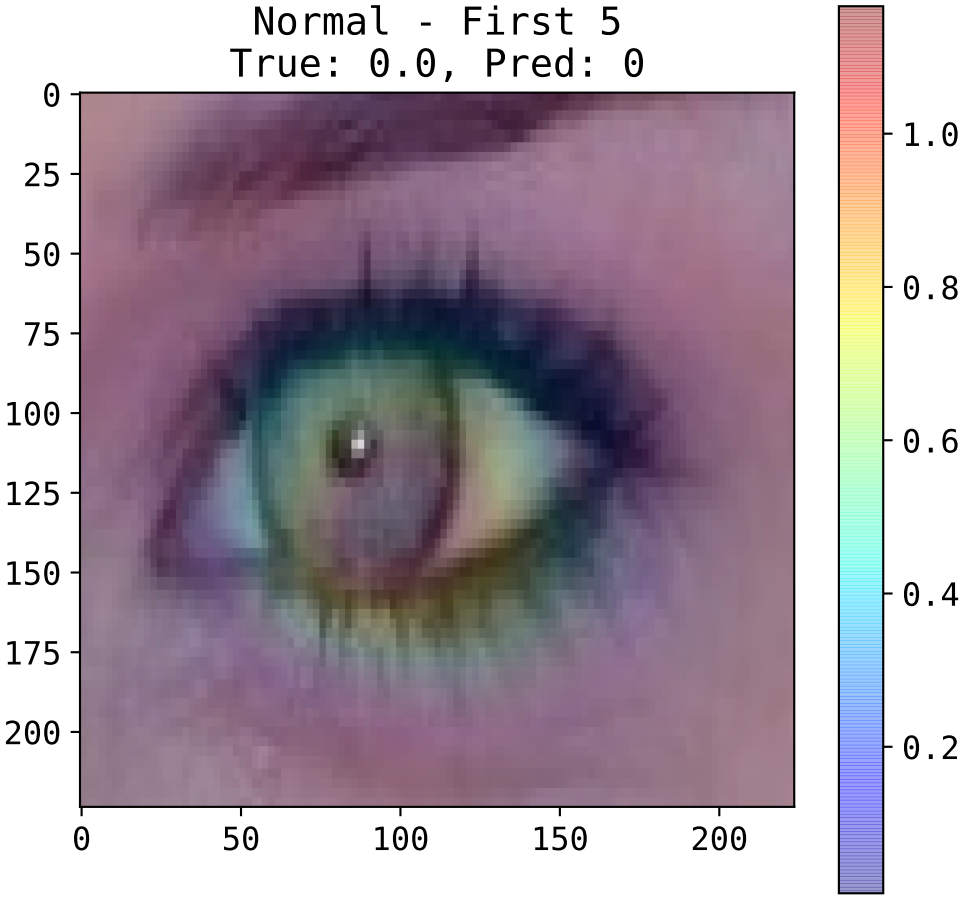

Precision Over Epochs (Two-Phase Training)
Precision
Total Epochs
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0
5
10
15
20
25
Train Precision
Val Precision
Fine-Tuning Starts

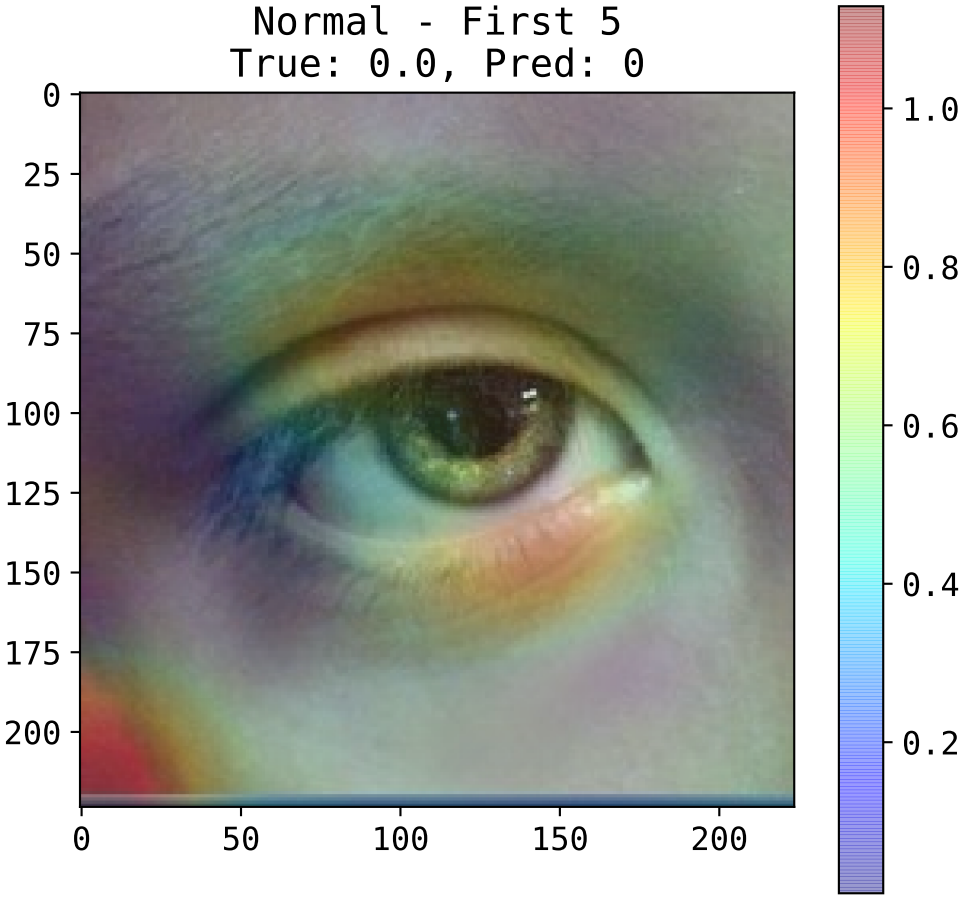

Receiver Operating Characteristic (ROC) Curve
Sensitivity (True Positive Rate)
1 - Specificity (False Positive Rate)
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (area = 0.96)

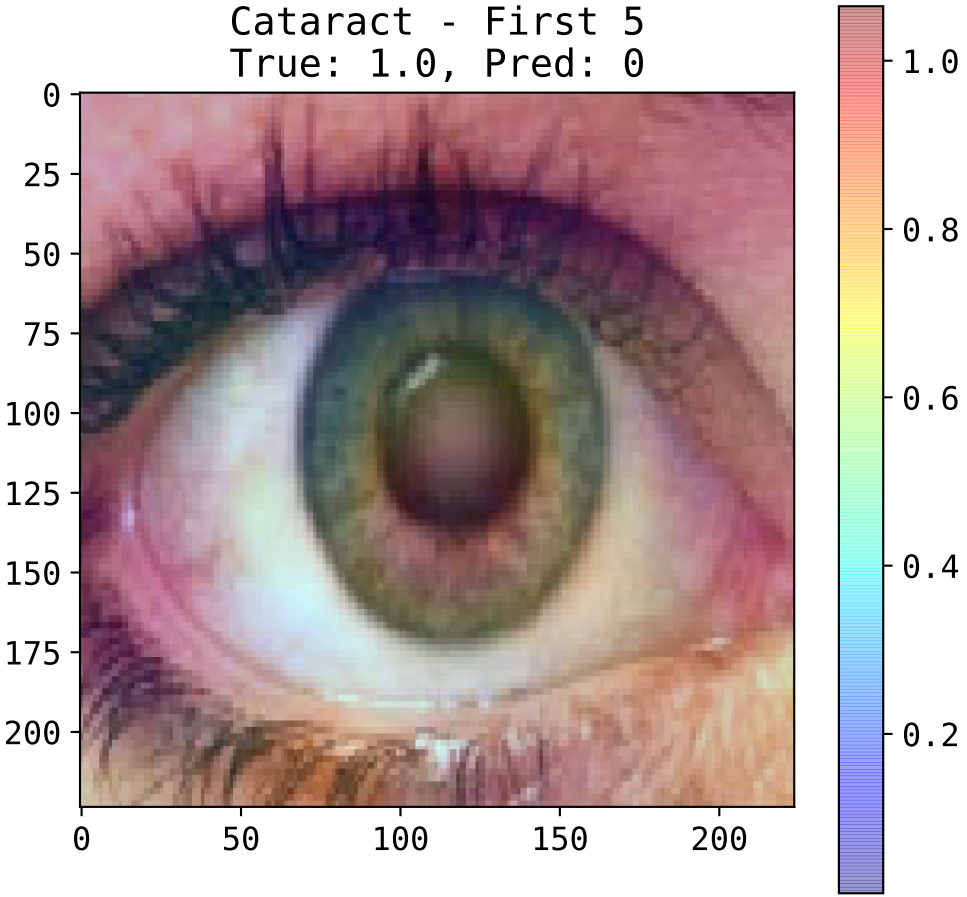

Normal - First 5
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2

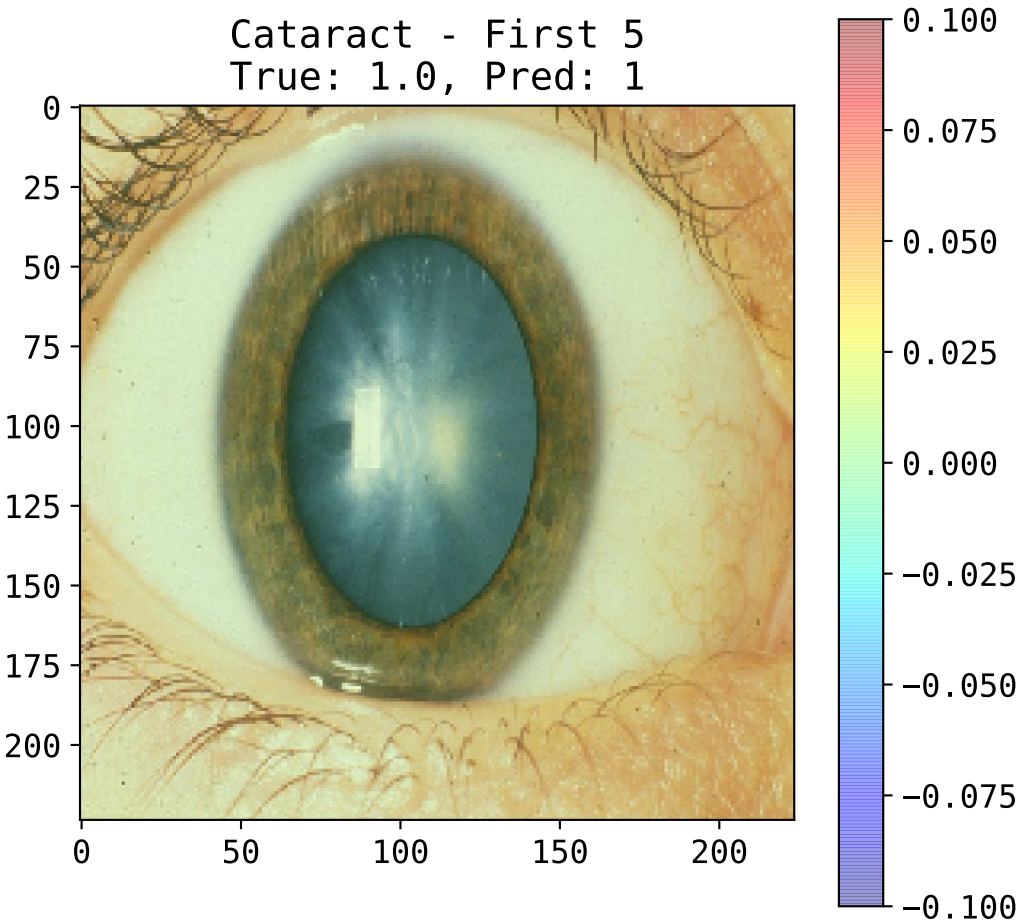

Normal - First 5
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2

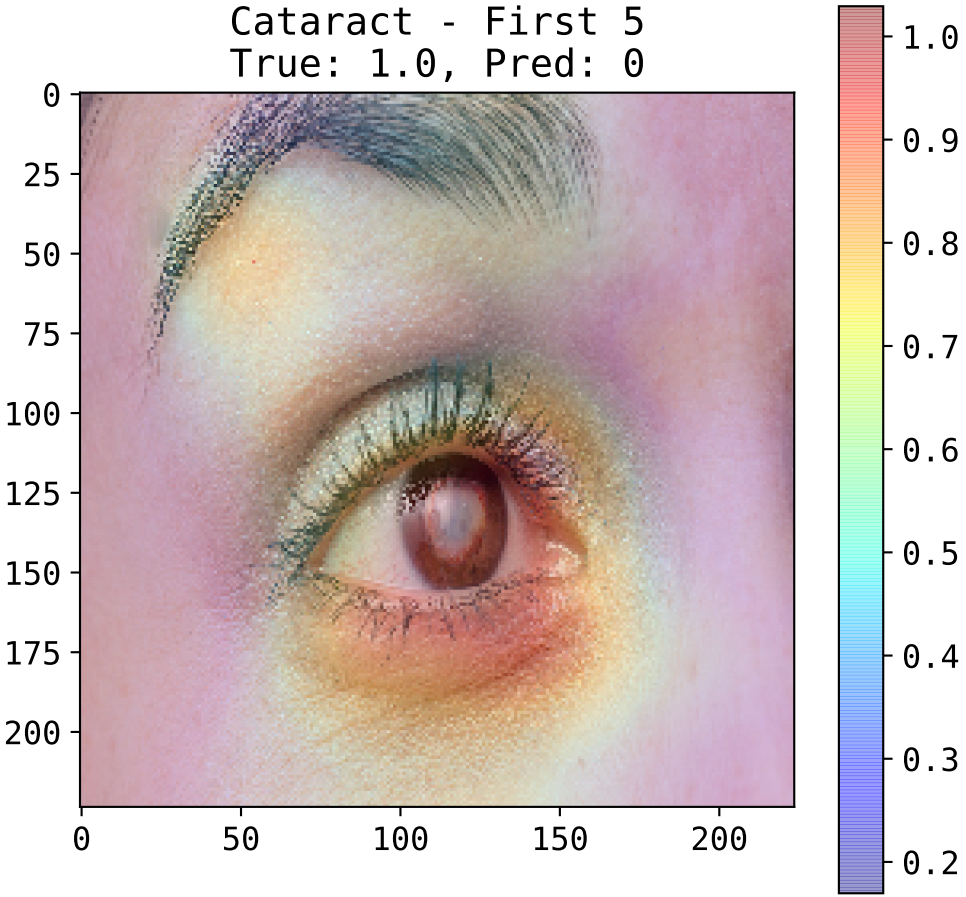

Normal - First 5
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2

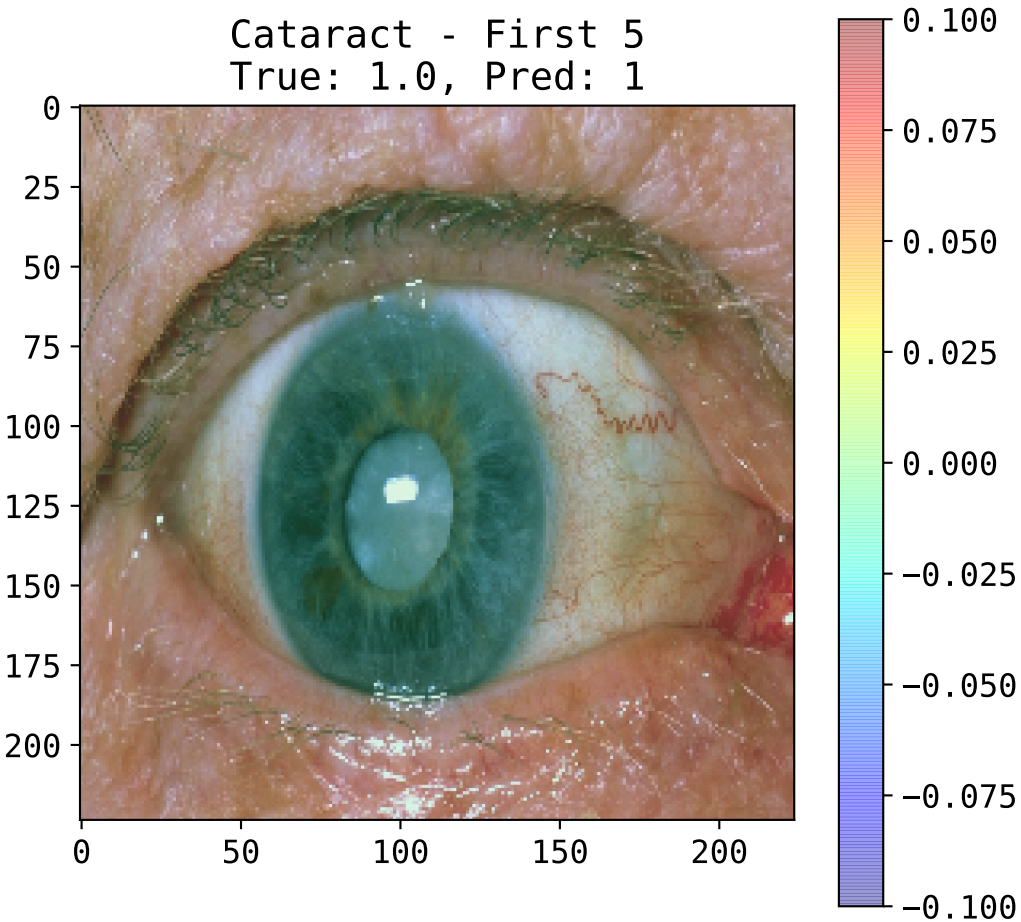

Normal - First 5
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2

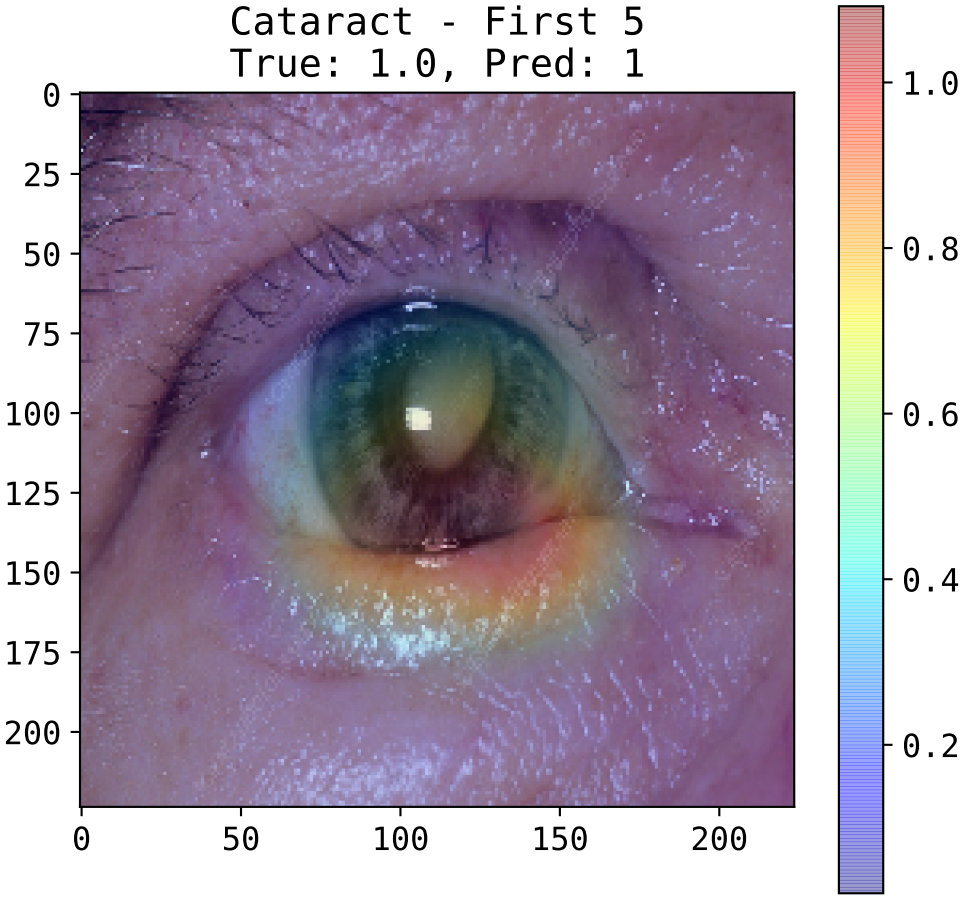

Normal - First 5
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2


Cataract - First 5
True: 1.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2


Cataract - First 5
True: 1.0, Pred: 1
0
25
50
75
100
125
150
175
200
0
50
100
150
200
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100


Cataract - First 5
True: 1.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2


Cataract - First 5
True: 1.0, Pred: 1
0
25
50
75
100
125
150
175
200
0
50
100
150
200
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100


Cataract - First 5
True: 1.0, Pred: 1
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2